

INGEN DYNAMICS INC.

Origami AI · Physical Intelligence Platform (PIC 2.0)

INTERNSHIP PROGRAM PLAN

Applied AI Analyst Intern · 8-Week Remote Program · v2.0

Intern	Xirui (Crissy) Chen
Position	Applied AI Analyst Intern
Engagement Type	Internship (Fully Remote, 8 weeks)
Time Zone	US Eastern (EDT, UTC-4) · InGen supervisor: IST (UTC+5:30)
Program Dates	June 15, 2026 – August 8, 2026
Start Date	June 15, 2026
Supervisor	Iqbal Patel — HR / Administration, InGen Dynamics Inc.
Coordination	Iqbal · HR & Administration
Contact	xiruichen0715@gmail.com (919) 600-1872
Academic Context	Tied to NYU CDS internship course for academic credit. Enrollment form signed at kickoff; mid-point and final evaluation rubrics (§6) satisfy the school's periodic evaluation requirement.
Document Version	v2.0 · Prepared May 2026 · Confidential — Internal Use

Referred by Jiahui · Document Version 2.0

1. Program Overview

1.1 Purpose

This plan structures an 8-week, fully-remote Applied AI Analyst internship for Xirui Chen at InGen Dynamics. The program bends Xirui's quantitative and time-series strengths toward physical AI: robot fleet telemetry analytics, anomaly detection, reinforcement-learning benchmarking, decision-model evaluation, and competitive intelligence on the embodied AI ecosystem.

Consistent with plans produced for Hank Zhong (R&D), William Chen (Data Analyst), Ziheng Wang (Data/BA), and Scarlett Zhang (Legal/CI), the entire program is self-contained on public data and open-source tooling. Xirui will not require access to any InGen internal system. All deliverables are personal GitHub repository artifacts that contain no InGen confidential content.

This is version 2.0 of the plan, prepared to reflect the most current résumé (including the TradingBase.AI role), the academic-credit requirement from NYU CDS, and the request for convenient interview and onboarding timing from the US East Coast.

1.2 Why This Program Fits Xirui's Profile

The program has been calibrated against Xirui's most recent résumé. Four observations shaped the design:

- High-cadence factor engineering on noisy time-series streams: the TradingBase.AI role (20+ market-microstructure factors over multi-chain ledger data; 5–10% anomalous-record filtering; statistical modeling with clustering and anomaly detection) is the closest possible analog to robot fleet telemetry analytics. Week 3's synthetic Aido Rover telemetry generator and Week 4's anomaly-detection benchmark are direct applications of this skillset to a new domain.
- Backtesting and KPI-driven strategy evaluation: Icarus Fund work (ARIMA / VARMA / state-space econometrics; Sharpe ratio, max drawdown, A/B-style comparisons) maps cleanly onto evaluating RL policies and decision-model quality — the same scientific muscles applied to a new substrate. Weeks 5–6 use this discipline explicitly.
- Large-scale data engineering and pipeline automation: the Caitong Securities role (1 TB+ financial datasets, automated analytical reports in SQL, automated trading-strategy lifecycle pipelines) signals that Xirui can build reproducible, end-to-end pipelines rather than one-off notebooks. This is a non-negotiable standard for all InGen intern deliverables.
- Stretch areas — embodied AI domain and RL: Xirui's résumé does not show prior robotics or RL work. Phase A (Weeks 1–2) and Phase C (Weeks 5–6) are specifically designed to develop that literacy at a credible analyst depth, using the statistical rigor already demonstrated in financial contexts.

1.3 Academic Credit — NYU CDS Compliance

This internship is tied to Xirui's NYU Center for Data Science internship course for academic credit. The following provisions apply throughout the program:

- Enrollment form: to be signed by Iqbal Patel (or designated supervisor) during the kickoff call on Day 1 of Week 1 and returned to Xirui for submission to the NYU academic coordinator within 5 business days of the start date.
- Periodic evaluation rubric: the mid-point evaluation (end of Week 4) and the final evaluation (end of Week 8) in §6 of this plan serve as the periodic evaluation artifacts required by the university. Both are completed jointly by the supervisor and intern, signed by both parties, and a copy is provided to Xirui for her academic record.
- Interview scheduling: the referral request specifies US East Coast time availability. All synchronous calls — including the pre-offer interview — should be scheduled in EDT morning slots (09:00–12:00 EDT / 18:30–21:30 IST) to accommodate the 9.5-hour offset.

- No modifications to the core program scope are required to satisfy academic credit. The existing deliverable cadence, rubric, and checkpoint structure meet standard NYU CDS internship-course requirements.

1.4 Operating Constraints

- No access to InGen internal systems, customer data, financials, or proprietary technical documentation. All work uses public datasets, open-source tooling, and InGen's published product portfolio.
- All deliverables must be safe for publication to a personal GitHub repository. No InGen confidential content may appear in any artifact.
- Open-source and free-tier toolchain only. See §8 for the full approved list.
- Every quantitative claim must carry a source citation or be explicitly labeled as synthetic/measured, with the test configuration stated.
- All files follow WXX_<TopicSnakeCase>.<ext> naming; weekly recaps follow weekly/Wk-NN-Recap.md.

2. Program Arc

Eight weeks organized as four two-week phases. Each phase produces written deliverables that feed the next phase; the four phases compound into the Week-8 capstone.

Phase	Weeks	Theme	Core Question	Outputs → Capstone
A	1–2	Orientation & Competitive Intelligence	What is InGen's position in the embodied AI ecosystem, and where are the strategic gaps?	Physical AI landscape brief; competitive matrix; strategic gap memo.
B	3–4	Telemetry Analytics & Anomaly Detection	Which anomaly-detection method best fits an Aido Rover fleet health monitoring use case?	Synthetic telemetry generator; 4-method benchmark; mid-point review.
C	5–6	RL Benchmarks & Decision-Model Evaluation	What do PPO/SAC baselines reveal about HTD-IRL needs, and how reliable is an AMDC eval harness?	RL benchmark runs; 50-scenario YAML harness; judge-reliability analysis.
D	7–8	Decision-Support Dashboard & Capstone	Can the outputs of Phases B–C be synthesized into a stakeholder-facing dashboard and a coherent 15–20 page report?	Streamlit dashboard; capstone report; executive deck; final evaluation.

2.1 Product Anchoring

Every weekly module names a primary InGen product anchor so technical exercises connect to a real platform context rather than running in the abstract. The five InGen platforms and PIC 2.0 are referenced as follows:

Product	Weeks	Edge AI / Analytics Relevance
Fari (eldercare companion)	Wks 6	Conversational decision evaluation — AMDC mode-switching scenarios for companion vignettes.
Senpai (educational robot)	Wks 6	Adaptive branching in AMDC harness — tutoring session scenario bank.
Sentinel Prime AI	Wks 4, 7	Anomaly-detection framing (Wk 4); operational dashboard view (Wk

(security)		7).
Aido Rover (outdoor patrol)	Wks 3–4, 7	Primary telemetry anchor; fleet health dashboard view (Wk 7); patrol-route RL scenario (Wk 6).
Aido Humanoid (bipedal research)	Wk 2	Competitive humanoid platform comparison in the vendor matrix.
Origami / PIC 2.0	Wks 1, 5–6	Foundation-model orientation (Wk 1); HTD-IRL and CRL-MRS RL benchmarks (Wk 5); AMDC evaluation (Wk 6).

3. Week-by-Week Plan

Each week is structured around a single objective, concrete deliverables with WXX_ naming conventions, a named InGen product anchor, tools and data sources, and a Friday self-check the intern uses before requesting weekly sign-off.

Week 1 · Jun 15 – Jun 20, 2026 · Orientation: Physical AI Landscape & Toolchain Setup	
Product Anchor	Origami / PIC 2.0 (foundation-model orientation); cross-portfolio overview
Objective	Internalize the physical / embodied AI landscape, InGen's published product portfolio, and the PIC 2.0 conceptual framework. Stand up the local toolchain and push the first GitHub deliverable.
Tasks	- Read and take structured notes on five anchor references: one current VLA paper (e.g., OpenVLA or RT-2); one humanoid foundation-model paper (e.g., Figure Helix or 1X); one hierarchical RL survey; the IFR World Robotics 2025 executive summary (free public abstract); and one peer-reviewed sim-to-real survey. - Build a one-page conceptual map of the six PIC 2.0 foundation models (GRPO, STUM, SEOM, AMDC, HTD-IRL, CRL-MRS) using only public InGen materials plus a short literature review of the equivalent open-literature concepts. Install the toolchain: Python 3.11, JupyterLab, pandas, NumPy, scikit-learn, PyTorch, Gymnasium, Stable-Baselines3, PettingZoo, PyOD, plotly, Streamlit. Commit a W01_env_check.ipynb that imports all dependencies without error. - Stand up the GitHub repository with the agreed structure (§7). Push all Week 1 deliverables by Friday EOD EDT. Kickoff call with Iqbal Patel: walk through the plan, agree product-anchor priorities, sign the NYU enrollment form, confirm working norms.
Deliverables	- W01_PhysicalAI_Landscape_Brief.md — 3–5 page synthesis of the five readings plus an ecosystem map. - W01_PIC20_Conceptual_Map.md — one-page conceptual map of the six PIC 2.0 foundation models. - W01_env_check.ipynb — environment verification notebook committed to GitHub. - weekly/Wk-01-Recap.md — 300–500 word recap.
Tools & Resources	arXiv, IFR public reports, vendor websites (InGen, Figure, 1X, Agility, Boston Dynamics, Unitree, Sanctuary AI), Hugging Face, GitHub.
Self-Check	Can I articulate in three sentences each what GRPO, STUM, SEOM, AMDC, HTD-IRL, and CRL-MRS plausibly do, and which open-literature concept each maps to?

- Does my env_check notebook run end-to-end on a fresh clone without errors?
- Is the NYU enrollment form signed and in Xirui's hands for submission?

Week 2 · Jun 22 – Jun 27, 2026 · Competitive Intelligence: Physical AI Vendors & Foundation Models

Product Anchor	Aido Humanoid (research-platform comparison); all five platforms cross-referenced
Objective	Produce a defensible competitive matrix of foundation-model and embodied-AI vendors, cross-referenced against InGen's published portfolio, and identify three strategic gaps where InGen's positioning is differentiated.
Tasks	• Profile at least 10 vendors across humanoid, service-robot, and outdoor-patrol categories: e.g., Figure, 1X, Agility Robotics, Boston Dynamics, Unitree, Sanctuary AI, Aptronik, Galbot, UBTech, Tesla Optimus. For each: positioning, disclosed foundation-model architecture, hardware platform, target verticals, funding stage, and recent press milestones. • Maintain explicit source citations for every cell of the matrix. No uncited numeric claims. • Cross-reference InGen's five public platforms (Fari, Senpai, Sentinel Prime AI, Aido Rover, Aido Humanoid) against the matrix. For each InGen platform, identify the single closest comparable and state the one dimension on which InGen appears differentiated. • Identify three specific strategic gaps in the competitor field that InGen's portfolio is positioned to address. Write the gap analysis as a 3-page memo — clause-level drafting, quantitative data, source citations. The rigor expectation mirrors the standard set in earlier InGen intern evaluations.
Deliverables	• W02_Competitive_Matrix.xlsx — 10+ vendors × ~10 fields with cited sources; one tab per cluster (humanoid, service, patrol). • W02_Strategic_Gap_Memo.md — 3-page memo: 3 differentiated InGen positions, each with a quantitative anchor and a source citation. • weekly/Wk-02-Recap.md.
Tools & Resources	SEC EDGAR, Crunchbase free tier, vendor websites, IFR public abstracts, arXiv, Google Patents, press releases.
Self-Check	• If I had 90 seconds to brief Iqbal Patel on the single most important strategic gap, which one would it be, and what is the one quantitative data point that anchors it? • Is every numeric cell in my matrix traceable to a public, datable source? • Do my three gap arguments each state an explicit assumption that could be falsified by a competitor announcement?

Week 3 · Jun 29 – Jul 4, 2026 · Synthetic Robot Telemetry — Generation & Exploratory Analysis

Product Anchor	Aido Rover (outdoor patrol fleet)
Objective	Translate prior factor-engineering and time-series experience to robot fleet telemetry. Build a reproducible synthetic generator that simulates an Aido Rover patrol fleet, then perform statistically rigorous EDA and engineer 15+ features.
Tasks	• Design and implement a synthetic telemetry generator: 50 simulated Aido Rover units over a 30-day horizon at 1 Hz, producing joint angles, IMU (3-axis accel + gyro), motor current per drive, battery voltage and SoC, wifi RSSI, GPS fix quality, and discrete task-success flags.

	• Model realistic distributions: motor current with thermal drift, battery SoC with daily charge cycles, RSSI with location-dependent multipath, and 1–2% baseline rate of benign measurement noise. Apply the same distributional discipline used on TradingBase ledger streams. • EDA: ADF stationarity tests, STL seasonality decomposition, pairwise correlation structure, missingness pattern analysis. • Engineer 15+ features documented in a feature dictionary with rationale, computation formula, expected distribution, and known failure modes. Include short-window rolling statistics, cross-sensor consistency ratios, and inter-unit z-scores. • Parameterize the generator with a CLI (seed, fleet-size, horizon); confirm bit-for-bit reproducibility from seed alone.
Deliverables	• W03_Telemetry_Generator.py — parameterized, seedable, with docstrings and a CLI. • W03_Telemetry_EDA.ipynb — EDA notebook with stationarity tests, correlation heatmaps, missingness analysis. • W03_Feature_Dictionary.md — 15+ features, each with rationale, computation, and a distribution plot. • weekly/Wk-03-Recap.md.
Tools & Resources	NumPy, pandas, scipy.stats, statsmodels (ADF, STL), plotly, seaborn.
Self-Check	• Could a second engineer regenerate my dataset bit-for-bit from my code, seed, and config alone? • Does my feature dictionary survive a code review by someone who has never seen this codebase? • Are my distributional choices (drift rate, noise level) documented with an explicit rationale?

Week 4 · Jul 6 – Jul 11, 2026 · Anomaly Detection on Fleet Telemetry · Mid-Point Review

Product Anchor	Sentinel Prime AI (anomaly framing) · Aido Rover (telemetry substrate)
Objective	Build, benchmark, and recommend a multi-method anomaly detection pipeline for robot fleet health monitoring, with statistically defensible method comparison. Present findings at the mid-point review.
Tasks	• Inject 5 controlled, well-documented fault patterns into the Week 3 dataset: motor stall, gradual sensor drift, accelerated battery degradation, GPS jitter (spoofing-like), and intermittent software hang. • Implement and benchmark four methods: Isolation Forest, One-Class SVM, Local Outlier Factor (LOF), and an LSTM autoencoder. Compute precision, recall, F1, and AUROC per fault type. • Run each method across 10 seeded train/test splits; report mean $\pm$ std. Run a paired Wilcoxon signed-rank test on the AUROC differences. This is the same discipline applied to backtested strategy comparisons at Icarus Fund — significance testing, not single-run comparisons. • Write a one-page recommendation memo: which method is preferred under which operating regime (latency-constrained vs. accuracy-maximizing), with explicit caveats. • Prepare a 6–8 slide mid-point summary deck for the formal review with Iqbal Patel.
Deliverables	• W04_Anomaly_Benchmark.ipynb — full benchmark with leakage controls, result tables, and Wilcoxon test.

	- W04_Method_Recommendation_Memo.md — 1-page recommendation with explicit assumptions and limitations. - W04_MidPoint_Deck.pptx — 6–8 slides summarizing Phases A–B. - Mid-point evaluation rubric (§6) completed and signed jointly with supervisor. - weekly/Wk-04-Recap.md.
Tools & Resources	PyOD, scikit-learn, PyTorch (LSTM autoencoder), scipy.stats (Wilcoxon), MAPIE (optional conformal calibration).
Self-Check	- If the benchmark numbers are tight between two methods, what is the operational reason I would still pick one over the other for a Sentinel Prime AI deployment? - Are my fault-injection patterns well-documented enough that another engineer could reproduce the exact fault conditions? - Does my recommendation memo state an explicit accuracy threshold below which the deployment would be unacceptable?

Week 5 · Jul 13 – Jul 18, 2026 · RL Benchmarking — Hierarchical & Multi-Agent Concepts

Product Anchor	PIC 2.0 — HTD-IRL (hierarchical task decomposition / inverse RL) and CRL-MRS (continual RL / multi-robot scaling)
Objective	Develop working analyst familiarity with the literature foundations behind HTD-IRL and CRL-MRS by running small, well-controlled RL benchmarks and producing a memo with transferable lessons for a multi-robot systems designer.
Tasks	- Run baseline PPO and SAC on two Gymnasium environments: LunarLanderContinuous-v3 and BipedalWalker-v3. Track learning curves over 5 seeds × 2 algorithms × 2 environments. Record convergence stability, wall-clock cost, and sample efficiency. - Run one multi-agent benchmark in PettingZoo (e.g., simple_spread) with PPO; briefly inspect coordination behavior. - Identify three transferable lessons that a hierarchical or multi-robot policy designer would care about, framed against what HTD-IRL and CRL-MRS conceptually need to deliver. Keep claims honest: this is a literature-anchored analyst exercise, not a research contribution. Distinguish clearly what the experiments demonstrated from what the literature claims, and explicitly avoid overgeneralizing from toy environments to fielded robots.
Deliverables	- W05_RL_Benchmarks/ — training scripts, TensorBoard logs, seeded result CSVs, learning-curve plots. - W05_RL_Analyst_Memo.md — 4-page memo with three transferable lessons and explicit citations to the underlying literature. - weekly/Wk-05-Recap.md.
Tools & Resources	Gymnasium (pip), Stable-Baselines3, PettingZoo, TensorBoard, seaborn; arXiv for HTD-IRL and CRL-MRS literature.
Self-Check	- Have I been careful to distinguish what the experiments demonstrated from what the literature claims? - Are my three transferable lessons framed from the perspective of someone designing a multi-robot system, not just summarizing benchmark numbers? - Do my seeded results reproduce cleanly on a fresh clone?

Week 6 · Jul 20 – Jul 25, 2026 · Decision-Model Evaluation Harness (AMDC Framing)

Product Anchor	PIC 2.0 — AMDC · Senpai (tutoring scenarios) · Fari (companion vignettes) · Aido Rover (patrol-route arbitration)
Objective	Build a lightweight, reproducible evaluation harness for AMDC-style adaptive decision behavior on simulated scenarios, with documented inter-rater reliability methodology.
Tasks	• Author a 50-scenario YAML test bank covering: priority arbitration (Aido Rover patrol-route selection under energy constraints), mode switching (Sentinel Prime AI alert escalation under bandwidth limits), and conversational decision branching (Senpai tutoring session and Fari companion vignettes). • Each scenario specifies inputs, allowed actions, and a target behavior (correct action label). • Implement a baseline rule-based controller and a baseline LLM-judged scoring rubric using a small open Hugging Face model. • Score each scenario across three seeded judge runs. Compute Krippendorff's alpha (or Cohen's kappa across judge seeds); discuss what the alpha value implies about the usefulness and reliability of the harness. • Document the harness with clear contracts, schema, and README so a future InGen engineer can extend the scenario bank without reading the implementation details.
Deliverables	• W06_Decision_Eval_Harness/ — scenarios/, scorer/, results/, README.md with run instructions. • W06_Eval_Methodology.md — 4-page methodology note: harness design, scoring rubric, reliability analysis, and known limitations. • weekly/Wk-06-Recap.md.
Tools & Resources	PyYAML, Hugging Face transformers (small open model, e.g., Flan-T5-small or similar), pandas, statsmodels, krippendorff (pip install krippendorff).
Self-Check	• If a colleague said 'your harness rewards the wrong thing,' what is the single most defensible thing I can say about why my scoring rubric is reasonable? • What is the most honest thing I can say about where the harness would fail? • Does my Krippendorff alpha reflect the correct level of agreement for three seeded judges on 50 scenarios?

Week 7 · Jul 27 – Aug 1, 2026 · Decision-Support Dashboard

Product Anchor	Sentinel Prime AI (operational view) · Aido Rover (fleet health view)
Objective	Synthesize the Weeks 3–6 outputs into a single analyst-facing Streamlit dashboard usable by three distinct personas: engineering manager, product manager, and customer success.
Tasks	• Build a Streamlit dashboard (preferred for portability) or Power BI Desktop dashboard. Cover three views: fleet health (anomaly detector results from Week 4), policy benchmark (RL convergence summaries from Week 5), and decision-eval scorecards (judge-rated AMDC scenarios from Week 6). • Design explicitly for three personas: engineering manager (needs precision/recall at a glance), product manager (needs trend and platform comparison), customer success (needs alert status per fleet unit). Document which control surface serves which persona. • Include a one-command launch script so a reviewer can spin up the dashboard from a clean clone in under 5 minutes. • Record a 5-minute narrated screencast walking through each view and each persona's use case.

Deliverables	- W07_Dashboard/ — app code, sample data, requirements.txt, and a one-command launch script. - W07_Dashboard_Design_Doc.md — per-persona design rationale plus annotated screenshots. - W07_Dashboard_Walkthrough.mp4 (or unlisted Loom link) — 5-minute narrated screencast. - weekly/Wk-07-Recap.md.
Tools & Resources	Streamlit (preferred) or Power BI Desktop; plotly, pandas; Loom or OBS Studio for screen recording.
Self-Check	- Have I designed for a viewer who has 60 seconds and one question, or have I designed for myself? - Can a reviewer with no prior knowledge launch the dashboard in under 5 minutes from a fresh clone? - Does each dashboard view answer a decision question for the named persona, not just display data?

Week 8 · Aug 3 – Aug 8, 2026 · Capstone — Applied AI Analyst Report & Final Evaluation

Product Anchor	Full InGen portfolio: Fari, Senpai, Sentinel Prime AI, Aido Rover, Aido Humanoid, PIC 2.0
Objective	Integrate the eight-week body of work into a 15–20 page capstone report and a 12-slide executive deck. Complete the final academic evaluation rubric and the NYU course evaluation form.
Tasks	Draft the capstone report. Structure: (i) executive summary — one page with the single most important conclusion; (ii) competitive landscape and strategic gaps; (iii) telemetry analytics framework; (iv) anomaly detection recommendation; (v) RL benchmark findings and transferable lessons; (vi) decision-model evaluation methodology and reliability; (vii) dashboard summary and forward-looking suggestions for each InGen platform. Build the 12-slide executive deck: title, agenda, competitive landscape snapshot, telemetry + anomaly results, RL lessons, decision-eval reliability summary, dashboard overview, per-platform recommendations, key limitations, and next-steps slide. - Record a 15-minute capstone presentation walkthrough. Treat the final review session as a real partner meeting — prepare for questions on methodology and product implications. - Complete the final academic evaluation rubric (§6) jointly with the supervisor. Submit the NYU internship-course evaluation form. - Tag a v1.0 release of the GitHub repository, ensuring every weekly artifact is reproducible from a clean clone. Write a one-page retrospective: most surprising finding, weakest section, and what a 12-week version would deepen. Shared verbatim — no polishing for the supervisor's eye.
Deliverables	- W08_Capstone_Report.docx — 15–20 page report with figures, tables, and citations. - W08_Capstone_Deck.pptx — 12-slide executive deck. - W08_Capstone_Presentation.mp4 — 15-minute recorded walkthrough. - Final evaluation rubric (§6) completed and signed; NYU internship-course evaluation form submitted. GitHub repository: v1.0 tagged; README complete with reproduction instructions.

	• weekly/Wk-08-Final-Recap.md.
Tools & Resources	Microsoft Word / Google Docs for report; PowerPoint / Google Slides for deck; GitHub Releases for v1.0 tag; Loom or OBS for recording.
Self-Check	• If this single repository were the only thing a future hiring manager saw, would it tell a clear story about what I can do? • Does my executive summary give a reader who skips the rest of the report the correct conclusion? • Is every number in the capstone traceable to a notebook cell, a cited source, or explicitly labeled as synthetic?

4. Deliverables Index

All deliverables live in Xirui's personal GitHub repository. File naming follows WXX_<TopicSnakeCase>.<ext>. Nothing in any repository is InGen-confidential. The mid-point and final rubrics in §6 are scored against this exact list.

Wk	Artifact	Description	Due	
1	W01_PhysicalAI_Landscape_Brief.md	3–5 page synthesis of 5 anchor readings + ecosystem map	Fri Jun 20	
1	W01_PIC20_Conceptual_Map.md	One-page map of 6 PIC 2.0 foundation models	Fri Jun 20	
1	W01_env_check.ipynb	Environment verification notebook	Fri Jun 20	
2	W02_Competitive_Matrix.xlsx	10+ vendors × ~10 fields with cited sources	Fri Jun 27	
2	W02_Strategic_Gap_Memo.md	3-page memo: 3 differentiated InGen positions	Fri Jun 27	
3	W03_Telemetry_Generator.py	Parameterized, seedable Aido Rover fleet simulator	Fri Jul 4	
3	W03_Telemetry_EDA.ipynb	EDA: stationarity, STL, correlation, missingness	Fri Jul 4	
3	W03_Feature_Dictionary.md	15+ engineered features with rationale and distribution plots	Fri Jul 4	
4	W04_Anomaly_Benchmark.ipynb	4-method benchmark with Wilcoxon significance tests	Fri Jul 11	
4	W04_Method_Recommendation_Memo.md	1-page recommendation memo with assumptions & caveats	Fri Jul 11	
4	Mid-point rubric (signed)	Joint evaluation form + NYU partial credit form (if applicable)	Fri Jul 11	
5	W05_RL_Benchmarks/	PPO / SAC / multi-agent runs; seeded CSVs, TensorBoard logs	Fri Jul 18	
5	W05_RL_Analyst_Memo.md	4-page memo: 3 transferable lessons, cited literature	Fri Jul 18	
6	W06_Decision_Eval_Harness/	50-scenario YAML harness; scorer; results; README	Fri Jul 25	
6	W06_Eval_Methodology.md	4-page methodology note with reliability analysis	Fri Jul 25	
7	W07_Dashboard/	Streamlit app: fleet health + RL bench + decision scorecards	Fri Aug 1	

7	W07_Dashboard_Design_Doc.md	Per-persona design rationale + screenshots	Fri Aug 1	
7	W07_Dashboard_Walkthrough.mp4	5-minute narrated screencast (or Loom link)	Fri Aug 1	
8	W08_Capstone_Report.docx	15–20 page capstone with figures, tables, citations	Fri Aug 8	
8	W08_Capstone_Deck.pptx	12-slide executive deck	Fri Aug 8	
8	W08_Capstone_Presentation.mp4	15-minute recorded walkthrough	Fri Aug 8	
8	Final rubric + NYU course form (signed)	Joint final evaluation; academic credit form submitted	Fri Aug 8	
1–8	weekly/Wk-NN-Recap.md	300–500 word weekly recap: built / blocked / learned	Each Fri	

5. Working Norms — Remote, Cross-Time-Zone

Xirui is in US Eastern (EDT, UTC–4) and the InGen supervising team is in IST (UTC+5:30) — a 9.5-hour offset. The program is substantially async-first, with one short scheduled overlap each week.

5.1 Checkpoint Cadence

Checkpoint	When	Format	Scope
Kickoff	Day 1, Wk 1	30-min call with Iqbal Patel	Walk through the plan; agree on product anchors; sign NYU enrollment form; confirm toolchain.
Weekly Sync	Each Friday	30 min + async recap	Deliverable review, blockers, next-week direction. Recap pushed to GitHub by EOD Friday 09:00 EDT.
Mid-Point Review	End of Wk 4	60-min formal review	Rubric scored across all 6 dimensions; written feedback; course corrections for Phases C–D; NYU partial progress form if required.
Final Review	End of Wk 8	60-min capstone + Q&A	Capstone presentation + Q&A; final rubric scored and signed; NYU internship-course evaluation form completed and submitted.

5.2 Daily & Weekly Rhythm

- Daily async update: short written end-of-day note in the agreed channel (Slack or email). Format: Yesterday / Today / Blockers. Two to four sentences.
- Weekly live check-in: Friday at 09:00 EDT / 18:30 IST. 30 minutes. Standing agenda: deliverable review, blockers, next-week scope. Async recap pushed to GitHub by EOD Friday EDT.
- Mid-point review: Friday of Week 4, 60 minutes. Joint rubric scoring and recalibration of Phases C–D scope if needed.
- Final review: Friday of Week 8, 60 minutes. Capstone presentation, Q&A, joint completion of the final rubric, and NYU evaluation form.
- Response-time norms: written messages within one supervisor business day. Nothing demanded out-of-hours given the 9.5-hour offset.

- Escalation: blockers held more than 24 hours should be raised with Iqbal and the HR/Administration team in cc.

5.3 Quality & Conduct Standards

- Substance over scaffolding: a deliverable that organizes a topic without analyzing it will be marked down. Every memo or notebook must contain named entities, quantified data, and source citations.
- Estimate is fine; uncited is not: if a fact cannot be sourced publicly, label it as an estimate and disclose the basis.
- No InGen confidential content: all work builds from public data, open-source tooling, and InGen's published portfolio. If in doubt, ask before committing.
- AI assistance is permitted: use of Copilot, Claude, or similar is allowed and encouraged for boilerplate. The README of any deliverable that uses AI assistance must include a one-line disclosure noting which sections were AI-assisted.

6. Evaluation Rubric — Mid-Point & Final

The same rubric is applied at the end of Week 4 (mid-point) and the end of Week 8 (final). Six dimensions, each scored 1–5, totaling 30 points. This rubric supports the NYU CDS internship-course academic-credit requirement: the completed form, signed by Xirui and the supervisor, is the artifact submitted to the academic coordinator at each evaluation point.

6.1 Scoring Scale

Score	Label	Definition
5	Exemplary	Substantively exceeds expectations. Could be handed to a senior engineer as a credible reference. Statistical claims are independently verifiable.
4	Strong	Meets expectations with margin; minor revisions only. Clear evidence-to-claim chain throughout.
3	Solid	Meets baseline expectations for a master's-level intern. Some gaps in rigor or polish but no major rework needed.
2	Developing	Below expectations on one or more dimensions. Specific corrective feedback documented; follow-up checkpoint scheduled.
1	Needs Intervention	Significant rework required. HR and Administration team awareness; formal development plan required.

6.2 Rubric Dimensions

Dimension	What Is Being Evaluated	Mid /5	Final /5	
Technical Depth & Rigor	Substantive technical content, not just structural scaffolding. Statistical claims have effect sizes, confidence intervals, and controlled comparisons.	__ / 5	__ / 5	
Analytical Reasoning & Statistical Discipline	Conclusions traceable to evidence. Limitations and assumptions stated explicitly. Uncertainty and operating regime clearly bounded.	__ / 5	__ / 5	
Code Quality, Reproducibility & Documentation	Clean clone + one command reproduces results. Seeds, configs, and data versions captured. Docstrings and READMEs sufficient.	__ / 5	__ / 5	
Written & Visual Communication	Writing concise and structured. Figures stand alone with self-contained captions. Audience (engineer vs. PM vs. CEO) clearly addressed.	__ / 5	__ / 5	
Initiative, Self-Direction & Remote Ownership	Blockers surfaced within one business day. Weekly deliverables land on time without prompting. Async communication clear and complete.	__ / 5	__ / 5	
InGen Product Anchoring	Weekly outputs explicitly tied to named InGen platform (Fari / Senpai / Sentinel Prime AI / Aido Rover / Aido Humanoid / PIC 2.0).	__ / 5	__ / 5	
TOTAL (out of 30)		__ / 30	__ / 30	

6.3 Per-Deliverable Feedback (Free-Text)

In addition to numeric scores, the supervisor provides three written notes per evaluation point: one strength, one specific development opportunity, and one concrete action for the next phase.

Note	Mid-Point (End of Week 4)	Final (End of Week 8)
Top Strength		
Development Opportunity		
Concrete Next Action		

7. Repository Structure & Reproduction

All work lives in Xirui's personal GitHub repository. Suggested name: xirui-chen-applied-ai-analyst. No InGen confidential content is committed. The supervisor's mid-point and final reviews are performed against a clean clone, on a fresh Python 3.11 virtual environment, following the README's one-command reproduction path.

Path	Contents
README.md	Program overview, constraint note, week-by-week deliverable index, one-command reproduction path.
requirements.txt	Pinned Python dependencies (output of pip freeze after env setup).
/week01_landscape/	Landscape brief, PIC 2.0 map, env check notebook.
/week02_competitive/	Competitive matrix XLSX, strategic gap memo.
/week03_telemetry/	generator.py, eda.ipynb, feature_dictionary.md.
/week04_anomaly/	Benchmark notebook, recommendation memo, mid-point deck.
/week05_rl/	Training scripts, TensorBoard logs, seeded CSVs, analyst memo.
/week06_decision_eval/	scenarios/, scorer/, results/, README.md, methodology note.
/week07_dashboard/	Streamlit app code, sample data, launch script, design doc, screencast.
/week08_capstone/	report.docx, deck.pptx, presentation.mp4.
/weekly/	Wk-NN-Recap.md files (one per week).
/docs/	This plan (PDF export), signed rubric scans.
/data/	Synthetic data only; gitignored beyond seeds and config; data/README.md explains how to reproduce.

Open-source and free-tier tools only. Consistent with the toolchain used across InGen's concurrent internship programs.

Category	Tools / Sources
Language & Core Analysis	Python 3.11, pandas, NumPy, scipy, statsmodels
Machine Learning	scikit-learn, PyTorch (LSTM autoencoder), PyOD (Isolation Forest, OC-SVM, LOF), MAPIE (optional conformal calibration)
Reinforcement Learning	Gymnasium, Stable-Baselines3 (PPO, SAC), PettingZoo (multi-agent), TensorBoard
Visualization & Dashboards	plotly, seaborn, Streamlit (dashboard) or Power BI Desktop
Evaluation & NLP	Hugging Face transformers (small open models — Flan-T5-small or equivalent), krippendorff, PyYAML
Time-Series & Signals	statsmodels (ADF, STL, ARIMA), scipy.stats
Public Data Sources	arXiv (cs.RO, cs.AI), IFR World Robotics public abstracts, SEC EDGAR, Crunchbase free tier, Hugging Face datasets, Kaggle, vendor websites
Productivity & Delivery	GitHub, JupyterLab, VS Code, Loom or OBS Studio (screen recording), draw.io / Excalidraw (diagrams)

9. Closing Note from the Supervisor

Eight weeks is a compressed runway, and Xirui arrives with a background that is unusually well-matched to the work. The TradingBase.AI factor-engineering pipeline, the Icarus backtesting discipline, and the Caitong large-scale data automation experience collectively represent exactly the quantitative rigor and production-mindset this program demands. What is genuinely new ground is the embodied AI domain and the RL substrate — Phase A and the opening of Phase C are structured to close that gap quickly.

The academic-credit requirement from NYU CDS is built into the cadence from Day 1: the enrollment form is signed at kickoff, and the mid-point and final evaluation rubrics in §6 satisfy the university's periodic evaluation requirement without adding administrative overhead to the technical program.

The program is async-first by design given the 9.5-hour EDT/IST offset, but the weekly Friday sync at 09:00 EDT / 18:30 IST gives a reliable overlap window. Interview scheduling should target this same window for consistency.

The single best predictor of a strong final deliverable is the same across all InGen intern programs: substantive depth over structural scaffolding. A notebook with a real AUROC number, an explicit fault-injection description, and a Wilcoxon test result beats a polished memo that hedges every claim. The evaluation rubric is calibrated accordingly.

Intern

Name: Xirui (Crissy) Chen

Signature: _____

Date: _____

Supervisor

Name: Iqbal Patel

Title: HR / Administration, InGen Dynamics Inc.

Signature: _____

Date: _____